

Statistical Algorithms Applied to Sports Ranking

Alibek Mamyrbay, Owen N Horton, Advisor: Andrew Sage

ABSTRACT

Whether they are immersed in a local or large-scale sports competition, many people desire to prove their team pride and demonstrate how they stack up among like groups. Many are involved in some way with sports fanaticism, making accurate ranking systems an increasingly important statistical subject. To understand and potentially improve statistical sports ranking, we researched datasets centered on numerous competitions using RStudio software. By exploring both a frequentist linear regression approach and the more time-sensitive Bayesian algorithm, we have come to various conclusions addressing the blend between statistical analysis and human interpretation.

Chess Rankings

BACKGROUND

- Players assigned ratings based on performance
- Competitions in International Chess Federation (FIDE)
- Data from *The Week in Chess*, providing weekly official game data
- 97,408 matches over 12 weeks

The structure of my dataset is shown below:

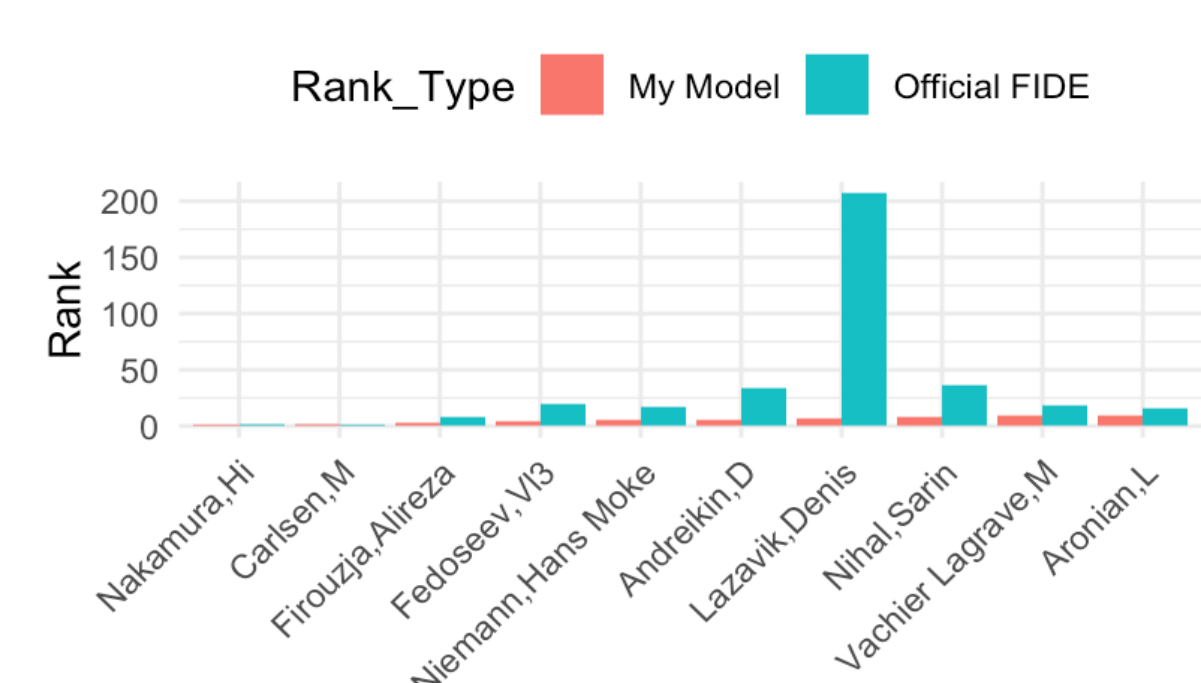
	Event	Date	White	Black	Result
1	13th Norway Chess 2025	2025-06-03	Erigaisi,Arjun	Caruana,F	1-0
2	13th Norway Chess 2025	2025-06-03	Wei Yi	Carlsen,M	1/2-1/2
3	13th Norway Chess 2025	2025-06-03	Nakamura,Hi	Gukesh,D	1-0
4	13th Norway Chess 2025	2025-06-05	Nakamura,Hi	Erigaisi,Arjun	1/2-1/2
5	13th Norway Chess 2025	2025-06-05	Gukesh,D	Wei Yi	1-0
6	13th Norway Chess 2025	2025-06-05	Carlsen,M	Caruana,F	1-0

The main **objective** of my research is to identify the model that best fits my data and provides reliable results.

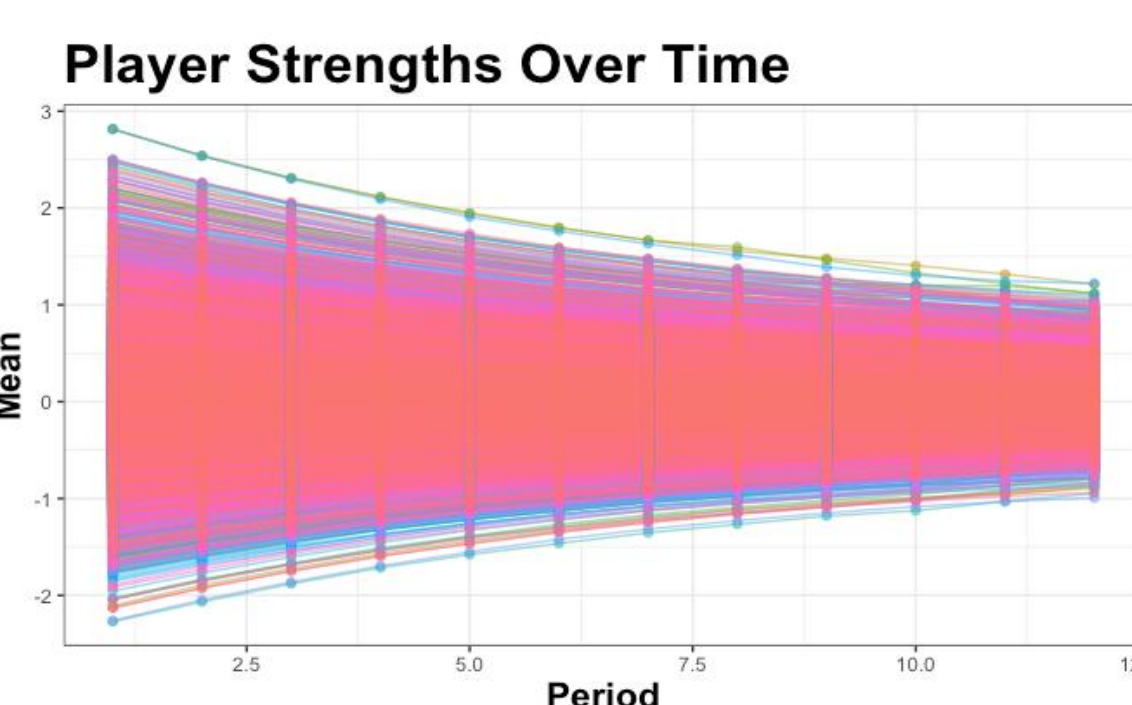
RESULTS

Best Model: Cauchy Model

- The model was run on a 12-week dataset from May 5 to July 21, 2025.
- The dynamic approach solved the core problem of players with few games ranking too high.
- **How it works:**
 - The 12 weeks of data were divided into 12 distinct time periods.
 - A unique strength rating is estimated for each player in each period, showing their performance level over time.
 - This structure requires sustained performance to achieve and maintain a high rank.
- The t-distribution allows the model to treat a single poor week as a rare event, not a substantial drop in skill.
- Overall, this model proved to be the most effective method for ranking active chess players among those I tested.



The bar plot illustrates the ten highest-ranked chess players according to my results based on the Cauchy model. Comparing the results from my model with the official FIDE rankings, we can clearly observe a strong correlation between the two. Since my dataset covered only the most recent 12 weeks, it has the potential to display current rankings even more accurately than the official data.



There are 11886 unique players in a 12-week dataset. There are also 12 time periods (weeks), so, in total, we see 142632 data points on the graph. They all are getting closer to the average (0) over time, illustrating that the estimations become more accurate with time.

METHODS

To achieve my objectives, I experimented with different models.

Model 1: Ordinary Least Squares Regression

- A baseline model
- Players with few games ranking at the top

Model 2: Weighted Least Squares

- Weighting games by opponent strength
- The same issue from OLS regression persisted

Model 3: Bayesian Constant Strengths Model

- The results were identical to those of the WLS model

Model 4: Cauchy Model

- Tracks player strength as it changes over time
- Solved the "few games" problem

2022 World Cup Rankings

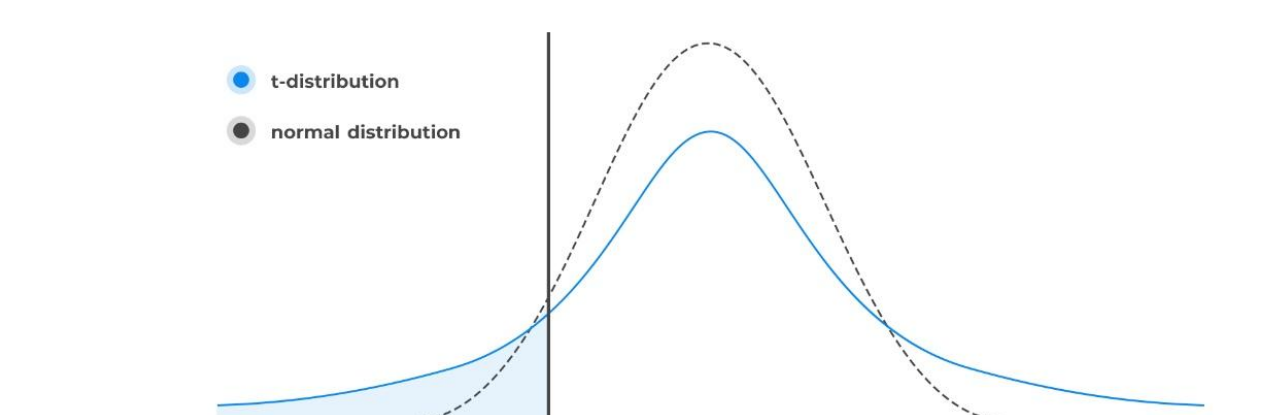
BACKGROUND

- Before a World Cup competition, teams are ranked using statistics and human discussion
- Extracted dataset contains 4,086 international soccer matches that took place in between the 2018 World Cup and the 2022 World Cup
- Statistical models experimented with include least squares regression, Bayesian statistics, and Cauchy Models
- The following parameters were deemed most important,
 - Goal Differential (Numeric Variable)
 - Home Field(1) or Neutral Field(0? (Binary Variable)
 - Friendly Match, Qualifier Match, or Cup Match? (Categorical Variable)
 - Date Played (Expressed as mm/dd/yyyy)

METHODS

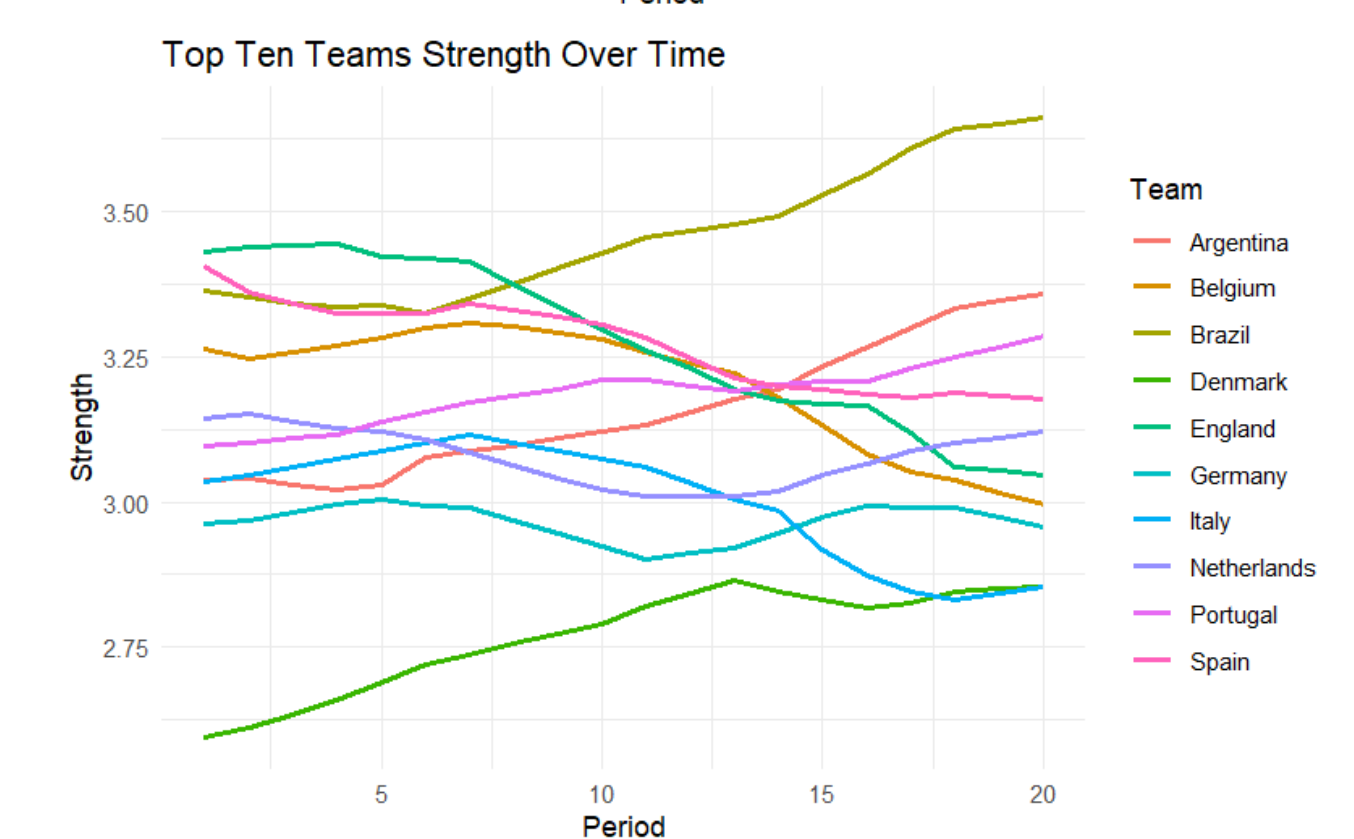
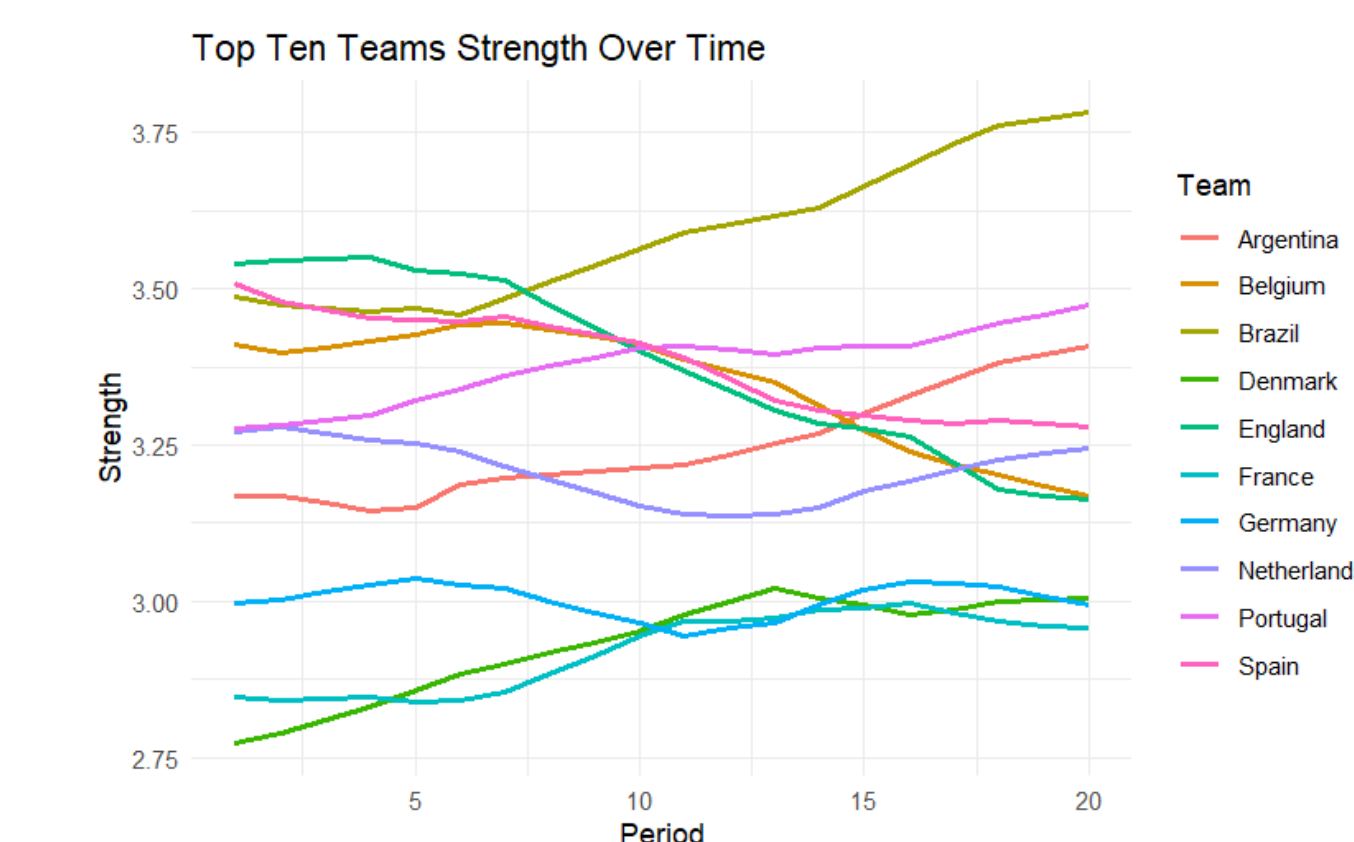
- Using R code, I imported the dataset into a standard least squares regression model; It proved to yield unreasonable results due to its lack of flexibility between weighted games
- Instead turned focus on Bayesian statistics and Cauchy models, for they are both reliant on prior assumptions (experimented with normal and T-distributions
- To account for time sensitivity, the nearly 4-year period within the dataset is split into 20 periods
- In probability, normal distributions place more weight on the mean and less on the tails, making it less flexible, while a T-distribution leaves more weight in the tails while maintaining the standard bell-curve shape
- In context, the normal distribution assumption causes the model to be more skeptical of good or bad periods, while T-distribution assumptions rewards or punishes good or bad periods to a greater degree

T-distribution vs Normal Distribution



Normal Distribution vs T-distribution; Source: Analyst Prep: <https://analystprep.com/cfa-level-1-exam/quantitative-methods/t-distribution-and-degrees-of-freedom/>

RESULTS



Top: Line graph of the top-ten finishing ranked teams based on the normal Cauchy model, split by 20 periods splitting four years of competitive matches. Bottom: A line graph of the top-ranked teams based on the T-distribution Cauchy model. Notice how the T-distribution assumption causes the changes of each team's predicted strength to be more pronounced.

- The following line graphs outline the Cauchy Model predicted changes in team strength for the top-ten predicted teams, with normal distribution assumptions on top and T-distribution assumptions on bottom. The Cauchy models appears to be the most statistically accurate ranking system relevant to the 2022 World Cup
- Both Cauchy model rankings performed better than the true 2022 World Cup rankings. The T-distribution performed slightly better in relative accuracy than the normal distribution. However, all of this is subject to random noise.
- The normal distribution assumption and T-distribution assumption yield slightly different results, and there is no clear-cut way to distinguish the superiority of either in any given context. This depends on human perspective
- Shows that human interpretation and subjective analysis is still viable, but perhaps should be relied on less than it is currently

CONCLUSIONS

Combining the results drawn from our two projects, we can conclude that the Cauchy model proved to be the most effective in generating accurate rankings. This indicates that accounting for change over time is especially germane in determining a team or player's strength. Despite brief experimentation with both normal and T distribution assumptions, we are unable to guarantee the legitimacy or superiority of either due to the random noise of sports competitions. However, we can generalize that these assumptions should be applied to a time-sensitive Cauchy model, effectively optimizing the current league rankings. By addressing the quantifiable and measurable variables while keeping our human assumptions open and flexible, we can more consistently distinguish changes in strength within a competition.